

Automatic Image Translation of Long Ancient Egyptian Texts for Augmented Reality Applications

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Figure 1: Example use of the proposed pipeline for recognition and translation in AR applications.

ABSTRACT

We introduce a novel long-texts-adapted open-source pipeline¹ for optical characters recognition and translation of Ancient Egyptian hieroglyphic texts. Designed for various materials of inscriptions and long inputs, our system opens the way for the creation of augmented reality applications for researchers and enthusiasts. Evaluated on a newly constructed exhaustive dataset, the system demonstrates a solid chrF++ score of 64 points.

Index Terms: OCR, image translation, augmented reality

1 INTRODUCTION

Ancient Egypt is one of the most enduring civilizations in the world. To this day, it excites the hearts of both scientists and ordinary people and left behind many fascinating monuments, including written ones. While interpreting these texts remains challenging, the key difference between Ancient Egyptian texts and modern ones is their form: not Unicode, but steles, tombstones, papyri, and their images. This highlights the need for complex augmented reality (AR) tools at the intersection of optical character recognition (OCR) and machine translation (MT) that go beyond simple glyph recognition and provide fast and accurate live interpretation.

Despite growing interest in digital humanities, there is a lack of open-source systems capable of processing Ancient Egyptian text, not to mention mobile applications. Moreover, while ancient texts

are often lengthy, modern MT models [2] have not been tested on such extended sequences.

We introduce an open-source pipeline capable of processing diverse inputs, from clean digital images to noisy photographs, by integrating context-aware image-to-glyphs recognition and translation of long texts. Designed for real-world variability and linguistic complexity, our system bridges visual and linguistic understanding, and could be a great backbone for AR applications.

2 APPROACH

Our end-to-end pipeline (see Fig. 1) consists of two sequential components. The first one is a context-aware OCR module that detects lines of hieroglyphs directly from images and predicts them as Unicode symbols, enabling robust recognition across diverse inputs.

The second stage is an Ancient Egyptians-to-German and English MT model, trained on close-to-reality longer coherent texts. Unlike previous approaches that treat translation as a phrase-level task, our model is trained to understand and generate fluent, context-sensitive translations of several consecutive clauses. This improves the interpretation of complex Ancient Egyptian texts.

For models training, we transformed and extended the Thesaurus Linguae Aegyptiae (TLA) [1] dataset.

2.1 OCR

In [2], the authors presented the OCR pipeline based on simple models and consisting of four parts: line and hieroglyph detection, and hieroglyph classification and sorting. It is capable of long-line processing, but the detection module encounters difficulties, resulting in reduced accuracy and reliability. As well, the method does not take advantage of contextual information during glyph classification, despite the context-dependent nature of writing.

Thus, we present the pipeline consisting of only line detection and TrOCR, adapted for wide images. TrOCR is an effective model, which leverages the Transformer architecture for both image understanding and text generation. To fine-tune it, we used carefully

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¹<https://github.com/AIRI-Institute/akhet>

annotated, diverse, and based on the TLA real-image dataset of 87 images and 984 clauses; the mean line size is 35.6 hieroglyphs. We also generated two synthetic datasets (word- and sentence-level), using the method presented in [3]. An example lines of real and generated data are shown in Fig. 2.



Figure 2: Samples of the real (up) and synthetic (down) data.

The methods comparison showed that the TrOCR-based pipeline is indeed better for all types of data (see Tab. 1). In addition, the TrOCR-based approach significantly improves speed, which is of great importance for resource-constrained devices applications.

Table 1: Performance comparison for OCR tasks on synthetic and real data. Labels "word" and "sent" stand for word- and sentence-level synthetic datasets correspondingly, "real" stands for real datasets, and CER stands for character error rate. Speed is given in images per second. Best scores are marked in bold.

model/task	CER _{word}	CER _{sent}	CER _{real}	speed
approach [2]	17.9	22.8	30.1	0.54
TrOCR _{small}	7.9	5.0	27.4	4.6
TrOCR _{base}	6.3	3.6	21.7	3.4
TrOCR _{large}	6.7	3.0	19.6	3.1

2.2 Translation

The translation dataset comes from TLA 2018 dump and contains approximately 100,000 parallel clauses in Ancient Egyptian (different stages), English and German: 28,457 from hieroglyphs to German, 3,197 to English, 88,569 from transliteration to German, and 2,330 to English. Ancient Egyptian texts are available in two forms: hieroglyphs and transliteration. The average text length is 273 hieroglyphs, while the average clause contains 22.6 ones.

To adapt translation models for the long texts, we created five nested datasets: the first one contains 1-clause samples (e.g. phrases), the second one – 2-clause samples etc., while the fifth contains long 5-clause samples. Then, each dataset was used for models testing. The 1-clause dataset and the combination of 1-, 2-, and 3-clause datasets was used as a training and validation dataset. Considering the low-resource status of the Ancient Egyptian and the latest practices, we fine-tuned *facebook m2m100.418M* and *nllb-200-distilled-1.3B* models, following the scheme described above.

To evaluate the fine-tuned models we exploited four metrics: BLEU, chrF++, ROUGE and METEOR. Fig. 3 clearly shows that models trained in a common 1-clause setup handle long samples from the test set much worse than adapted 1-, 2-, 3-clause models: the decrease reaches 20 BLEU and 14 chrF++ points. Best models obtain the scores of 53 and 64 chrF++ (hieroglyphic and transliterated input correspondingly), which indicates good translation quality and solid real-use perspectives.

3 CONCLUSION AND FUTURE WORK

In this work, we presented a complete, fast, and long-text adapted pipeline for the image translation of Ancient Egyptian hieroglyphic texts. The system includes OCR and translation modules, and demonstrates promising results in automated evaluation of both speed and quality. Looking ahead, we plan to develop an interactive interface that integrates the entire pipeline.

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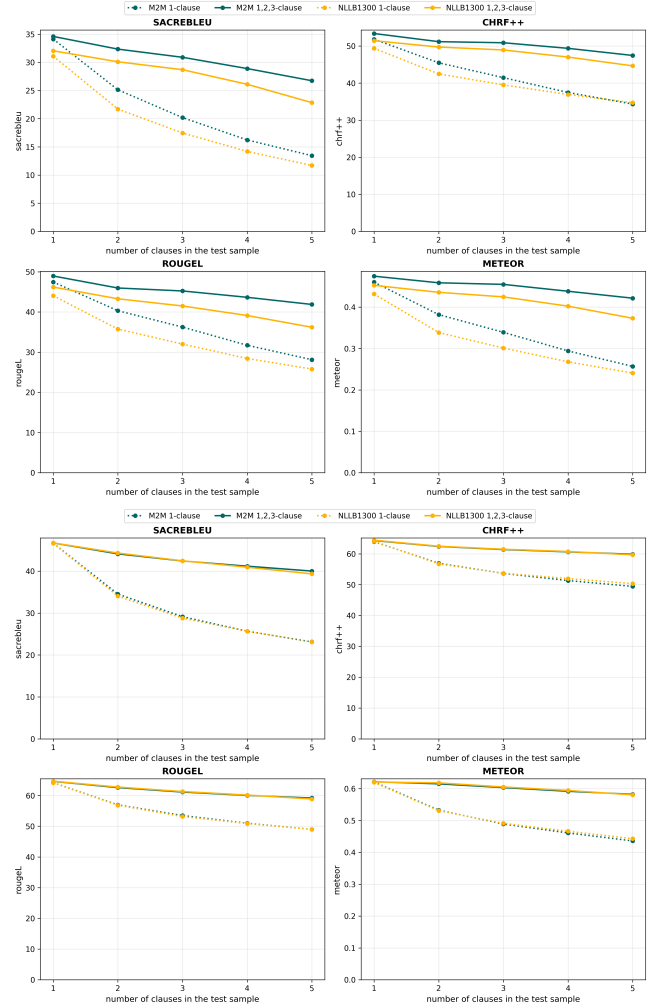


Figure 3: Metrics for M2M (green) and NLLB (yellow) models, trained on 1-clause (dotted) and on 1-, 2-, 3-clause (solid) combined datasets. Hieroglyphs (up) and Transliteration (down) to German.

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REFERENCES

- [1] Berlin-Brandenburgische Akademie der Wissenschaften and Sächsische Akademie der Wissenschaften zu Leipzig. Thesaurus linguae aegyptiae. Web app version 2.2.1.1, Corpus issue 19, 2025. Edited by Tonio Sebastian Richter and Daniel A. Werning (on behalf of the BBAW), and Hans-Werner Fischer-Elfert and Peter Dils (on behalf of the SAW). 1
- [2] M. Golyadkin, I. Humonen, Y. Plevokas, E. Bureeva, E. Alexandrova, and I. Makarov. Automatic interpretation of ancient egyptian texts for education and research. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters, SIGGRAPH Posters '25*. Association for Computing Machinery, New York, NY, USA, 2025. doi: 10.1145/3721250.3743025 1, 2
- [3] M. Golyadkin, V. A. Rubanova, A. Utkov, D. Nikolotov, and I. Makarov. Evaluation of egyptian hieroglyph classification across diverse writing styles. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 1–8, 2025. doi: 10.1145/3746027.3754797 2